A Dominant Polynomial Parametrization of $xyz + t^2 + 1 = 0$

Abstract

We construct an explicit three-parameter polynomial family of integral solutions to $xyz+t^2+1 = 0$. The resulting polynomial map $\mathbb{A}^3_{\mathbb{Q}} \to V(xyz + t^2 + 1)$ is dominant, so the family is Zariski dense. A one-parameter specialization in which all four coordinates are nonzero is also recorded.

1 The polynomial construction

Lemma 1 (Factorization identity)**.** *For arbitrary elements U, V of any commutative ring,*

$$\big(U + V(U^2 + 1)\big)^2 + 1 = (U^2 + 1)\big((1 + UV)^2 + V^2\big). \tag{1}$$

Proof. Expanding the left-hand side and extracting the common factor $U^2 + 1$ gives

$$\begin{aligned}\big(U + V(U^2 + 1)\big)^2 + 1 &= U^2 + 1 + 2UV(U^2 + 1) + V^2(U^2 + 1)^2\\ &= (U^2 + 1)\big(1 + 2UV + V^2(U^2 + 1)\big)\\ &= (U^2 + 1)\big((1 + UV)^2 + V^2\big).\end{aligned}$$

□

Theorem 2 (Three-parameter integral family)**.** *Let $a, b, c \in \mathbb{Z}$, and put*

$$r = a + b(a^2 + 1). \tag{2}$$

Define

$$\begin{aligned}X &= -\big((1 + cr)^2 + c^2\big),\\ Y &= (1 + ab)^2 + b^2,\\ Z &= a^2 + 1,\\ T &= r + c(r^2 + 1).\end{aligned} \tag{3}$$

Then

$$XYZ + T^2 + 1 = 0. \tag{4}$$

Moreover, $X < 0$, $Y > 0$, and $Z > 0$; in particular, $XYZ \neq 0$. Thus (3) *defines a polynomial map $\mathbb{Z}^3 \to \mathbb{Z}^4$ whose image consists of integral points on* (4)*.*

Proof. Apply Lemma 1 first with $(U, V) = (a, b)$. By (2),

$$r^2 + 1 = (a^2 + 1)\big((1 + ab)^2 + b^2\big) = ZY. \tag{5}$$

Apply the same identity again with $(U, V) = (r, c)$. Using (5) and the definition of X, we obtain

$$\begin{aligned}T^2 + 1 &= (r^2 + 1)\big((1 + cr)^2 + c^2\big)\\ &= ZY(-X) = -XYZ,\end{aligned}$$

which is exactly (4).

Finally, $Z = a^2 + 1 \geq 1$. The integer Y is a sum of two squares and cannot vanish: if $Y = 0$, then $b = 0$ and $1 + ab = 0$, which would give $1 = 0$. Hence $Y \geq 1$. Similarly, $-X$ is a sum of two squares and cannot vanish: if it vanished, then $c = 0$ and $1 + cr = 0$, again giving $1 = 0$. Therefore $-X \geq 1$, so $X < 0$. □

The family is infinite, since its Z-coordinate, $a^2 + 1$, is unbounded as a ranges over $\mathbb{Z}$.

2 Dominance

Lemma 3 (Jacobian criterion used here)**.** *Let k be a field of characteristic zero and let $f_1, \dots, f_n \in k[u_1, \dots, u_n]$. If*

$$\det \left(\frac{\partial f_i}{\partial u_j} \right)_{1 \le i,j \le n}$$

is not the zero polynomial, then $f_1, \dots, f_n$ are algebraically independent over k. Equivalently, the polynomial map $(f_1, \dots, f_n) : \mathbb{A}_k^n \to \mathbb{A}_k^n$ is dominant.

Proof. Suppose, to the contrary, that a nonzero polynomial $P \in k[s_1, \dots, s_n]$ satisfies $P(f_1, \dots, f_n) = 0$. Choose such a P of least total degree. Differentiating this relation with respect to $u_1, \dots, u_n$ and applying the chain rule yields

$$(P_{s_1}(f), \dots, P_{s_n}(f)) \left(\frac{\partial f_i}{\partial u_j} \right)_{i,j} = 0.$$

The Jacobian matrix is invertible over the rational function field $k(u_1, \dots, u_n)$, because its determinant is nonzero. Consequently $P_{s_i}(f) = 0$ for every i. By the minimality of P, each nonzero partial derivative P_{s_i} is impossible, since it has smaller total degree. Hence every P_{s_i} is the zero polynomial. Characteristic zero then forces P to be constant, contradicting the choice of a nonzero relation. Thus the f_i are algebraically independent.

For the final assertion, the ideal of the Zariski closure of the image is precisely the kernel of the induced homomorphism $k[s_1, \dots, s_n] \to k[u_1, \dots, u_n]$, $s_i \mapsto f_i$. Algebraic independence says that this kernel is zero, so the closure is all of $\mathbb{A}_k^n$. □

Proposition 4 (Zariski density)**.** *Let*

$$H = V(xyz + t^2 + 1) \subset \mathbb{A}^4_{\mathbb{Q}}.$$

The polynomial map $\Phi : \mathbb{A}^3_{\mathbb{Q}} \to H$ defined by (2)–(3) is dominant. Hence the family in Theorem 2 is a dominant polynomial parametrization of H.

Proof. First, H is irreducible. Indeed, set $R = \mathbb{Q}[y, z, t]$ and regard

$$F = xyz + t^2 + 1 = (yz)x + (t^2 + 1)$$

as a polynomial in $R[x]$. If $F = AB$ were a nontrivial factorization, then $\deg_x F = 1$ would force one factor, say A, to lie in R. It would then divide both coefficients yz and $t^2 + 1$. These coefficients are coprime in R: every irreducible divisor of yz is associated to y or to z, and neither y nor z divides $t^2 + 1$. Thus A is a unit, a contradiction.

Consider the composite of Φ with projection to the (z, y, t) coordinates:

$$\psi = (Z, Y, T) : \mathbb{A}^3_{\mathbb{Q}} \longrightarrow \mathbb{A}^3_{\mathbb{Q}}.$$

Since Z and Y are independent of c, while $\partial T / \partial c = r^2 + 1 = ZY$, its Jacobian matrix has the form

$$\frac{\partial(Z, Y, T)}{\partial(a, b, c)} = \begin{pmatrix} 2a & 0 & 0 \\ 2b(1 + ab) & 2r & 0 \\ * & * & ZY \end{pmatrix}.$$

Here

$$\frac{\partial Y}{\partial b} = 2a(1 + ab) + 2b = 2\big(a + b(a^2 + 1)\big) = 2r.$$

Therefore

$$\det \frac{\partial(Z,Y,T)}{\partial(a,b,c)} = 4arZY, \tag{6}$$

which is not the zero polynomial. Lemma 3 shows that ψ is dominant.

Let $U = \{(z,y,t) : yz \neq 0\} \subset \mathbb{A}^3_{\mathbb{Q}}$ and $H^\circ = H \cap \{yz \neq 0\}$. Projection

$$\pi : H^\circ \longrightarrow U, \qquad (x,y,z,t) \longmapsto (z,y,t),$$

is an isomorphism, with inverse

$$(z,y,t) \longmapsto \left(-\frac{t^2+1}{yz},\, y,\, z,\, t\right).$$

On $\psi^{-1}(U)$, the identity already proved gives $\Phi = \pi^{-1} \circ \psi$. Since ψ has dense image, its image in U is dense in U; hence the image of Φ is dense in H°. The set H° is a nonempty open subset of the irreducible variety H (it contains $(-1,1,1,0)$), and is therefore dense in H. Thus Φ is dominant. Since $\mathbb{Z}^3$ is Zariski dense in $\mathbb{A}^3_{\mathbb{Q}}$, the integral set $\Phi(\mathbb{Z}^3)$ is itself Zariski dense in H. □

3 A one-parameter specialization

Corollary 5. *For every $n \in \mathbb{Z}$, set*

$$\begin{aligned} x_n &= -\big((n^2-n)^2+1\big), \\ y_n &= (n-1)^2+1, \\ z_n &= n^2+1, \\ t_n &= n^4-2n^3+2n^2-n+1. \end{aligned} \tag{7}$$

Then

$$x_n y_n z_n + t_n^2 + 1 = 0,$$

and all four coordinates are nonzero.

Proof. In Theorem 2, substitute $a = n$, $b = -1$, and $c = 1$. Then

$$r = n - (n^2+1) = -(n^2-n+1),$$

and direct simplification gives exactly (7). The asserted equation follows from Theorem 2. Its proof already gives $x_n < 0$, $y_n > 0$, and $z_n > 0$. If $q = n^2-n+1$, then $q \geq 1$ and $t_n = q^2-q+1 \geq 1$, so $t_n \neq 0$ as well. □

Equivalently, Corollary 5 supplies the polynomial identity

$$\big(n^4-2n^3+2n^2-n+1\big)^2+1 = \big((n^2-n)^2+1\big)\big((n-1)^2+1\big)(n^2+1).$$

For example, $n = 2$ gives $(x,y,z,t) = (-5,2,5,7)$.

Remark 6 (Scope). Dominance is a geometric density statement. No claim is made that (3) represents every integral point of H; it gives an explicit Zariski-dense polynomial family of such points.